



SAHAYAK

Offline. Vernacular. On your side.

A scam shield and benefits guide for people new to digital money: at the CSC counter and on their own phone, with no internet, in Hindi and English, by voice or touch. Built and measured.

Ideas for India 2026 · Inclusive Innovation for Bharat · Financial Inclusion

Roshan Raj · Ishmiit Singh

Banked, then left to be robbed

Hundreds of millions of first-time account holders now get OTPs, UPI requests and scheme messages they were never taught to read, and miss benefits they are owed.

₹22,495 crore

lost to financial cyber fraud in 2025, across 24 lakh complaints

₹4,005 crore

lost by senior citizens in 2025 (1,03,488 complaints)

1 in 5

families with a UPI user hit by UPI fraud in the last 3 years; 51% never reported it

55%

of people aged 60+ know the old-age pension exists; 29% of poor elders receive it

And every safety tool assumes the internet, a smartphone, English and the cloud: the four things the people most at risk are least likely to have, or trust.

At the counter, and on the phone at home

One small computer at a CSC or bank-agent counter. Phones join its Wi-Fi, which has no internet; after one visit, the same checks run on the phone itself.

"Is this a scam?"

Paste, speak, photograph or describe a message, a call or a UPI QR. A verdict in milliseconds, the three reasons, what to do now, and a ready 1930 complaint.

"What am I owed?"

A spoken interview (median 5 questions) across 12 central schemes, health first: what you can get, the papers, where to go, what to say, and a printed slip.

Then, on their own phone

At home, when the call actually comes: the same scam check and benefits interview, on the phone, with no node and no internet. Nothing leaves the phone.

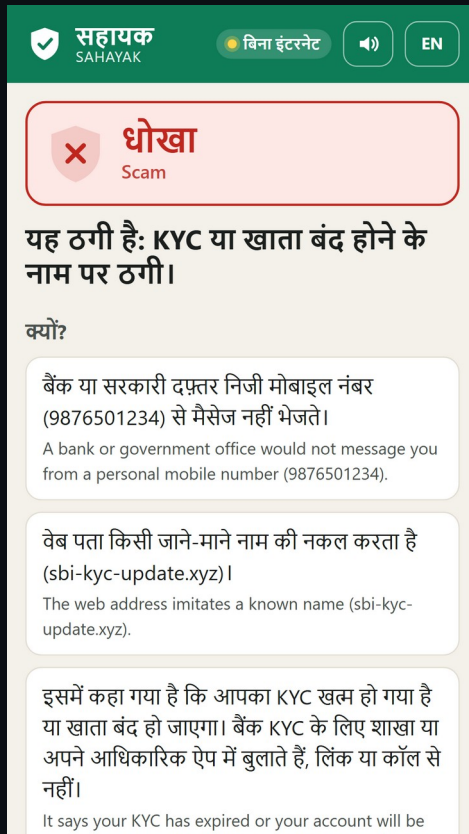


Try it now, on your own phone: roshworldwide.github.io/optum-v2

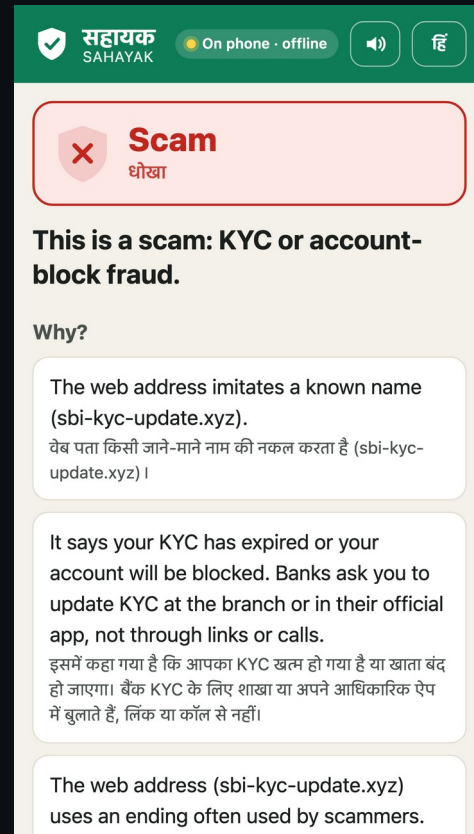
Check a message, switch on airplane mode, check another: it still works, and gives the node's exact answer.

● SEE IT WORKING

Built, and on a phone today



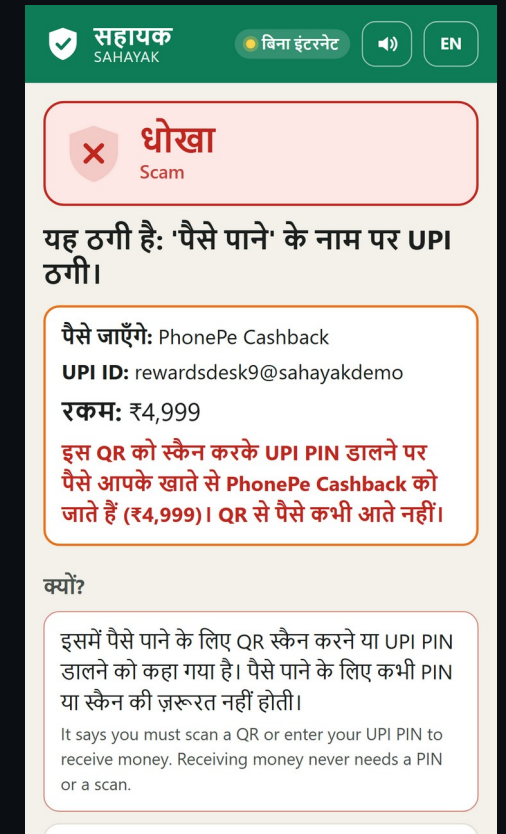
A fake KYC SMS, on the node



Airplane mode: checked on the phone



Age 74: health cover first



"Scan to get cashback" QR

Technology and implementation

- | | | |
|----|-------------------------------|---|
| 01 | Phone app | A web app, no install, Hindi and English, every screen read aloud. After one visit it works offline on the phone. |
| 02 | Fraud-Shield | 51 named signals in a signed content pack, a pattern matcher and a small classifier decide. Vetted templates explain; an optional local model may add a line, only through a safety gate. |
| 03 | Benefits Navigator | Scheme rules as data, each traced to its official page and date; yes / likely / unknown / no logic; asks only what can change the answer. |
| 04 | Offline voice | Speech in and out on the node (Vosk, Piper); amounts spoken as words; "I heard ..., is that right?" before any answer is used. |
| 05 | The node | One mini-PC or laptop at a CSC. Zero-egress monitor, Ed25519-signed packs updated by USB, operator console, captive portal. |
| 06 | Same answer everywhere | The phone runs the same rules from the same signed packs: identical results to the node on 1,453 messages and 7,983 interviews. |

What already exists, and what Sahayak adds

Already out there	What it does well	What Sahayak adds
Airtel and Jio network filters	Block spam calls and links at network scale; Airtel reports 68.7% lower fraud losses	Checks the message, QR code or call story in front of the person and says why, in Hindi, offline
Truecaller	Caller ID and spam lists for 339 million users	No account and no upload: nothing leaves the phone; covers UPI QR codes and screenshots
Google on-device scam alerts	On-device call analysis	Runs in any phone's browser; Google's works on Pixel 9+ only (under 1% of phones)
1930, cybercrime.gov.in, Sanchar Saathi	Reporting and blocking after the fraud	Catches it before the payment, and ends every scam verdict in a ready 1930 complaint
myScheme	4,700+ schemes in 15 languages, online, self-serve	Offline, spoken interview with a slip for the CSC; every rule and amount proven against official pages
Haqdarshak	₹19,233 crore in benefits through trained agents	The offline front door and the scam shield; a content partner, not a rival

Sources: Airtel, citing I4C data, Sep 2025; Truecaller, Q2 2026; TechCrunch, Nov 2025; myScheme.gov.in, Oct 2026; Haqdarshak impact analysis, 2025

It already works, and we measured it

91.9%

scams caught (blocklist: 51.4%)

2 ms

per verdict on a laptop CPU

15.4%

Hindi word error, real speakers

0.65 s

from mic release to spoken reply

100%

of 1,453 messages: the phone gives the node's exact answer

How we know, and what it does not show

ScamBench v0 frozen test split (60 messages, 95% CI 82.1%–100.0%); SchemeBench: 2,400 of 2,400 decisions match an independent re-derivation; Google FLEURS Hindi (418 recordings). Red team: 46 of 52 disguised scams caught on the first run (blocklist 33). v0 messages were written by our team, so those numbers are optimistic; real messages collected with consent come next.

₹5 lakh of health cover most seniors have not claimed

1.1 crore of ~6 crore

seniors aged 70+ had their Ayushman Vay Vandana card by Feb 2026

- Every benefits result leads with Ayushman Bharat; for anyone 70 or older it is the Vay Vandana card, ₹5 lakh of treatment a year whatever their income.
- One question, their age, finds them; the slip tells the operator what to make.
- The CSC makes the card at the counter, and is paid ₹20 per first-time card; the citizen pays nothing.
- The scam check flags "pay to get your Ayushman card" messages; the node counts seniors found, every month, with no personal data.

 सहायक
SAHAYAK

फ़ोन पर • बिना इंटरनेट

 EN

12 में से 2 योजनाएँ आपके काम की

उम्र: 74 साल हालात: इनमें से कोई नहीं

राशन कार्ड: BPL या प्राथमिकता (PHH) बचत खाता: हों

काम: अभी काम नहीं (गृहिणी, छात्र, सेवानिवृत्त)

पहले सेहत: मुफ्त इलाज

आयुष्मान भारत (PM-JAY) • मिल सकता है

आपके लिए ₹5 लाख का इलाज हर साल

काउंटर पर ऐसे कहें: "नमस्ते। मेरी उम्र 74 साल है। मुझे आयुष्मान वय वंदना कार्ड बनवाना है।"

आपको मिल सकता है 1

मिल सकता है

आयुष्मान भारत (PM-JAY)

आपके लिए ₹5 लाख का इलाज हर साल

सूचीबद्ध अस्पतालों में हर साल ₹5 लाख तक का मुफ्त इलाज।

कागज़, जगह और कारण

From "account opened" to "citizen safe"

Rupees at risk

The node adds up the money asked for in messages it flags, every month: never called "saved", always shown to the district without personal data.

People found, health first

People found eligible, by scheme, with seniors found for Ayushman Vay Vandana counted on their own; each slip lets the operator finish the form.

Reported in time

RBI compensation from 1 Jan 2027 needs a report within 5 days. Every scam verdict ends in a ready 1930 complaint, on the spot.

Pilot target, not a result

One district: about 20 villages and 50 CSC and bank-agent points, six months. Scams flagged and reported, people found eligible and cards made, all from the node's counters. A one-morning field test at a CSC (protocol and consent ready) comes first.

Grow by replication; paid for by those who gain

Replication, not reinvention

- 6 lakh functional CSCs, 4.7 lakh of them at gram-panchayat level: the counters already exist.
- A deployment is a node image, signed packs and a one-page operator guide; a new scam or scheme is a pack update by USB.
- Every phone that visits can keep its own copy, so protection spreads beyond the counter at no extra cost.
- No cloud and no per-query fee; open, tested rules any state can audit.

Who pays

₹25,000–35,000

once per node (an N150 mini-PC (8–16 GB) or a refurbished i5 laptop); each check costs nothing.

- Banks: from 1 Jan 2027 they share the cost of compensating fraud losses, so a loss prevented is money kept.
- The CSC operator: paid about ₹20 per first-time Ayushman card or e-Shram registration Sahayak sends to the counter; 86% of rural operators earn under ₹500 a month.
- The district: a signed monthly count of scams seen and people found eligible, to fund against.

Inclusive innovation on India's own rails

Financial inclusion, made safe

Protects the 59 crore Jan Dhan accounts and the UPI base the nation built, for the people who opened them last.

1930 and RBI's 2027 rules

Every scam verdict ends in the national helpline; fast reporting is what RBI's compensation now requires.

Ayushman Bharat

Finds seniors for the Vay Vandana card and sends them to the counter that makes it.

Digital India, last mile

Hindi and English by voice today, offline; the language data is a pack, so more languages follow.

DPDP Act

Privacy by design: consent first, nothing named, nothing written to disk, deleted by default.

CSC · BC · ASHA

Delivered through the public networks people already trust, with an operator at the counter.

Built in the open; seeking last-mile partners

CSC village-level entrepreneur or bank business correspondent

hosts the node and the field morning

seeking

A rural NGO or SHG federation

invites participants, checks the Hindi and the consent

seeking

A bank branch, regional office or SLBC

BC points for the pilot, real messages for ScamBench v1

seeking

Haqdarshak or myScheme

scheme content beyond 12 central schemes; Sahayak is the offline front door

seeking

Ready for each partner: a one-morning field protocol with consent in Hindi and English, a scoring script, and letters of intent. The code is MIT-licensed and open.

Builders who ship, asking for one district

Roshan Raj

Systems and AI-infrastructure engineer · Manipal Institute of Technology · 7 patents pending

Ishmiit Singh

Co-builder: product, content and evidence

Together: the node, the scam shield, the scheme navigator, offline voice, the phone engine, the operator console and the benchmarks behind every number in this deck.

The ask

A six-month, single-district pilot (about 20 villages, 50 counters, roughly ₹35-40 lakh) with:

- CSC e-Governance or the State IT department: the counters
- A bank BC network or SLBC: reach, and real messages for ScamBench v1
- NHA or the state health agency: Vay Vandana cards made at the counter
- A rural NGO: field testing and Hindi review

India banked half a billion people.

Sahayak helps keep them safe, and claim what they are owed.

At the counter and on their own phone · offline · in their language · nothing leaves the device · built and measured. Starting with ₹5 lakh of health cover for every senior who does not yet have the card.

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roshworldwide.github.io/optum-v2